

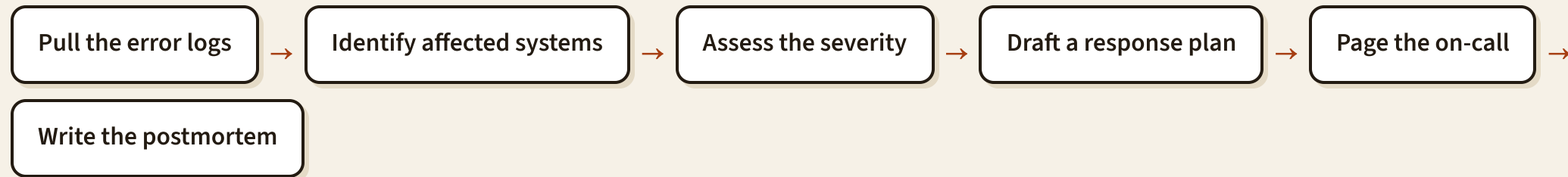
Architecture Over Intelligence

Designing reliable AI workflows

Amar Patel · The AI Fellowship Madrid · 9 Oct 2026



"Investigate this incident" is one job. *It isn't.*



One request sets off a chain of small jobs.

Each job needs the one before it. If one step drifts, every step after it drifts too.

The demo vs. production gap

In a demo

- One uninterrupted session
- Mistakes are annoying
- A human watches the whole time
- A frontier model

In production

- Compaction, reset, walk-aways
- Mistakes are data loss
- A human checks hours later
- A cheaper model that drifts

A demo works because nothing interrupts it.

In production, the session ends, the model changes, and a mistake now costs you.

A JSON file is the *source of truth*

● process-state.json		meeting_id: 2026-01-15_140000
current_step	2.6	
completed_steps	[-1, 0, 1, 2, 2.5, 2.6]	
user_confirmed.summary	true	
aborted	false	

Every step reads the same file before it runs. It shows the last step done — here 2.6, so the next is 3.

If the session dies, the next run starts from that same file and that same step, and redoes nothing.

A gate is *authority*, not just quality control



Auto

Runs without stopping. Nothing irreversible here, and a human checks it at a later gate.

Reads, health checks, sentiment.



Human

Stops and waits for your yes before continuing.

Anything that changes data.



Blocked

Cannot continue. Something required is missing or broken.

Missing file, corrupted state.

"Auto" does not mean the model got it right. It means a mistake here is cheap and always reviewed later.

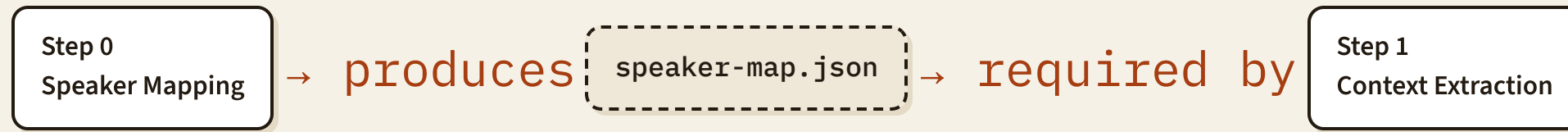


REVIEWS EVERY STEP

GUARDS ONLY THE IRREVERSIBLE

You don't remove the human as the model improves. You move it: from checking every step, to guarding only the steps that cannot be undone.

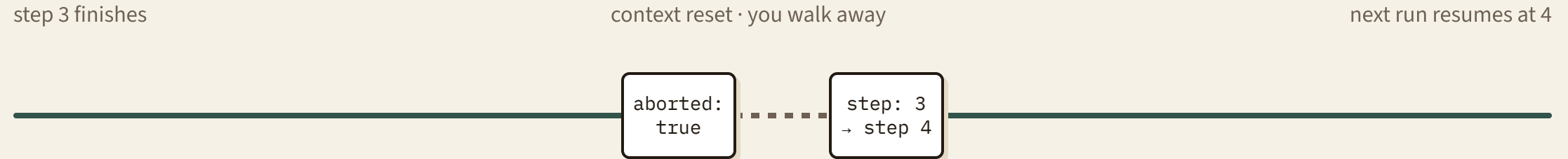
Each step declares its *contract*



Every step declares two things: what it produces, and what it needs.

Before a step runs, it checks its inputs really exist on disk. Missing input? It stops and says so — never guesses.

Interruption is not a bug



The file is written after every step, so nothing depends on memory.

Step 3 finishes → the file is on disk → the session dies (context reset, or you leave) → the next run reads the file → resumes at 4, redoing nothing.

One real pipeline: *11 steps, 8 LLM calls*



● stops for a human ● runs alone ○ optional

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HUMAN CHECKPOINTS

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SCRIPTS

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STATE FILE

Nine of the 11 steps stop for a human; two do not (sentiment runs alone, cleanup is optional).

What broke after months of daily use

I ran this every working day for months — processing my meetings end to end. Three things broke, and each fix pushed the decision into the file, not into the model.

- ✎ The state file got edited to the wrong step → *validate on load*
- ⏮ The LLM skipped a gate → *the file decides, not the prompt*
- ✓ The context reset mid-pipeline → *the file survived, and it resumed where it left off* — the win

Trust the file, and check the artifact itself.

Never trust what the previous step said it did.

The model is always changing ●

~~2024~~

A bigger model fixed the problem.

2026

Bigger models stopped helping. New models kept coming.

This month

Jev — it returns a decision, not a paragraph.

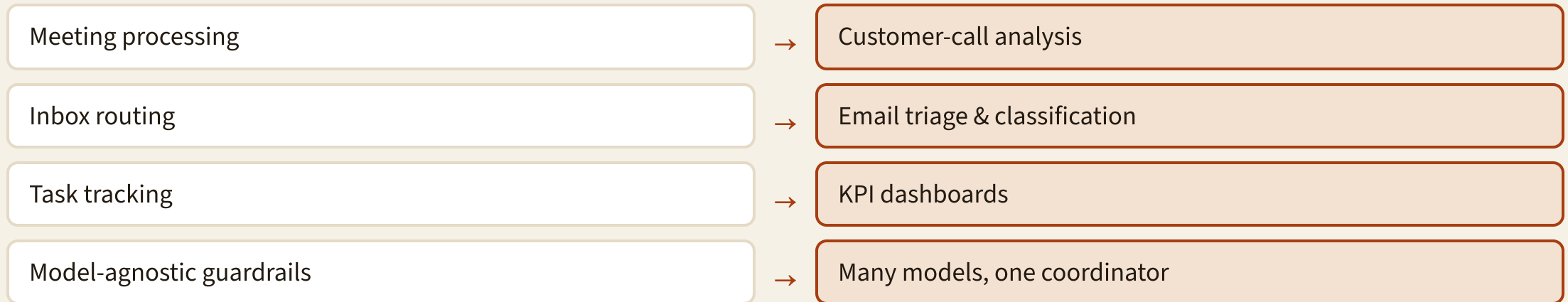
Scaling hit a wall: paying for a bigger model stopped buying reliability. Yet a new model still lands every month, so the pressure moves onto the cheaper ones — which is exactly where the drift lives.

This month it was Jev. It cannot hallucinate, because it does not write text. But the first good write-up of it was titled:

“The State Machine Is the Agent.”

And days later, researchers showed prompt injection can still bend its decisions. Even a model built to be safe sits behind a deterministic loop and a human.

The same pattern fits many jobs



Every row is the same shape: read → transform → a human approves → write to the system of record. Only the names change.

"Many models, one coordinator" just means several different models, each doing a step, all managed by the same state file.

Five rules — and how to check your own

1 ~~work~~ **state file** is the source of truth.

2 ~~work~~ **Gates** are structural, not optional.

3 Declare **artifacts** before they're needed.

4 Design for **interruption**.

5 **Architecture > intelligence.**

Run these three questions over your own agent:

a What is your single source of truth — a file, a row, a record?

b Where do you stop for a human, and can a step be skipped without one?

c Does each step check its inputs exist before it runs?

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Tonight: add one field — `current_step` — read it back before each step, and stop if it doesn't match

Key takeaways

- **The model changes. The structure stays.** A chain of steps breaks wherever it isn't supervised. Put a state file in charge.
- **The human does not disappear.** Better models move it to the irreversible steps. Consequence: the model cannot delete, overwrite, or send without a yes.
- **The same structure runs any model.** When next month's model arrives, your workflow does not change.

Reliability at the boundaries.